H524 Group Project Check-In: AI Divergence Experiment

Fall 2025

Dr. John Molitor — Due: See Canvas — 5% of Project Grade

Purpose: Discover how AI tools respond differently to biostatistical questions and learn to use prompt engineering to reach consensus.

Getting Your Dataset: Go to your group's Canvas page to download your assigned CSV file. Each group has been assigned one of five datasets (anorexia, toothgrowth, birthweight, pima diabetes, or diabetes trial). You'll upload this CSV to the AI tools in Step 2.

The Experiment (4 Steps)

Step 1: Propose a Research Question - Choose a **broad question about YOUR dataset** that might elicit different AI answers. Good questions ask about *method choice, assumptions, or interpretation.* Examples: "What statistical method should I use to analyze weight changes in the anorexia treatment data?" or "How should I compare tooth growth across supplement types and doses?"

Step 2: Consult Two Different AI Tools - Use the **EXACT same prompt** with two AI tools (Claude, ChatGPT, Copilot, or Gemini). Copy their full responses.

Step 3: Use Third AI as Judge - Ask a different AI: *"I asked two AI tools [your question] and got these responses: [paste both]. Rate their similarity 0-100 (0=different, 100=identical). Identify key agreements/disagreements."*

Step 4: Iterate to Increase Consensus - Try ≥2 modified prompts (add context, be specific, add constraints, request format). Re-score similarity each time.

Deliverable: 1-2 Page Report

Section 1: Team info (number, members, dataset)

Section 2: Your research question and why you expected divergence

Section 3: Experiment table showing ≥3 iterations with: Prompt text, AI 1 response summary, AI 2 response summary, Similarity score (0-100)

Section 4: Brief reflection (4-5 sentences): Did similarity increase? What prompt changes worked? What did you learn about AI variability? Which AI will you use for final project?

Grading (5 pts = 5% of Project)

Component	Pts
Question broad enough for divergence	1
Consulted 2 AIs with identical prompt	1
Used 3rd AI to score similarity	1
≥2 prompt iterations documented	1
Thoughtful reflection	1
Total	**5**

Why This Matters: Your final project (30% of grade) requires comparing AI tools critically. This teaches you that AI responses vary by tool AND prompt. Mastering prompt engineering now strengthens your analysis later.